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Using NLP and Supervised ML to Assess Real-Time Downside Risk in the Technology Industry During the U.S. - China Trade War

Abstract

This project develops a real-time political risk-monitoring framework for the U.S. China technology trade war by implementing natural language processing on policy event descriptions with a rolling 5% value at risk (VAR) on the Nasdaq 100. I convert a timeline of tariff events and export restrictions into daily sentiment and tech-industry relevance scores using VADER and keyword weighted TF-IDF, then aggregate them into 1 to 3 day rolling features. These text derived signals are aligned with a 21 day VAR series to capture tail-risk spikes triggered by trade policy escalations. I employ and compare two supervised tree-based regressors, Random Forest and Gradient Boosting, to quantify how clusters of negative, tech-focused escalations drive downside risk, with Gradient Boosting yielding tighter predictions. Although there is negative out of sample R², the SHAP analysis pins sentiment, event count, and tech focus as the risk drivers. This approach closes the gaps left by current monitoring methods like earnings calls. This prototype lays the groundwork for dynamic hedging and real-time risk alert systems, allowing market stakeholders to manage political tail risk in technology investments.

Relevant Background

The United States and Chinese governments have been engaged in a contentious trade war that has heightened in severity in 2025. There has been a large impact on the technology industry specifically. On August 9, 2022, the CHIPS Act was signed into law by former President Joe Biden with the goal of strengthening the U.S. semiconductor industry by increasing domestic production capacity and innovation.¹ A key part of that legislation prohibited recipients of funding from the bill from expanding manufacturing in China or other adversarial countries.² Later, in October 2022, the U.S. leveraged the Foreign Direct Product Rule to control technology exports between foreign countries and China, if those products used American technology or tools.³ This aimed to push China behind in the race to advance technologically.⁴ In September of

¹ Communications, N. W. (n.d.). Chips and science act: What is it? and why is it important?. NYU.
<https://www.nyu.edu/about/policies-guidelines-compliance/compliance-matters-newsletter/Fall-2022/chips-and-science-act--what-is-it--and-why-is-it-important-.html>

² Ibid.

³ Lee, J., & Nellis, S. (2022, October 2). Explainer: What is “FDPR” and why is the U.S. using it to cripple China’s tech sector?. Reuters.

<https://www.reuters.com/technology/what-is-fdpr-why-is-us-using-it-cripple-chinas-tech-sector-2022-10-07>
⁴ Ibid.

2023, the U.S. pressured other governments and technology firms to restrict exports to China of manufacturing tools for semiconductors.⁵ They later rolled back this policy in September of 2024.⁶ In April of 2025, the U.S. imposed a tariff of 34% on all Chinese imports.⁷ In response, China matched with a 34% tariff, export restrictions on rare earth metals, and sanctions on U.S. companies.⁸ The export restrictions on rare earth metals were aimed at impacting U.S. defense and technology firms, since China produces 90% of the world's supply of these rare earth metals.⁹ The Trump Administration responded with 104% tariffs on China, which responded with 84% tariffs.¹⁰ This continued to escalate back and forth until May 12, 2025, when the U.S. and China agreed to reduce tariffs and reach a tentative, temporary agreement to pause the trade war.¹¹

When the U.S. imposed a 34 percent tariff on Chinese imports in early April 2025, markets swung violently. On April 3, the Dow plunged nearly 1,700 points and the S&P 500 fell almost 5 percent, while Apple, Microsoft, and Tesla saw one-day drops of 3.7 percent, 0.6 percent, and 2.6 percent, respectively.¹² After the May 12 tariff truce, the Nasdaq jumped 3.34 percent, Apple rose 6.2 percent, and Nvidia gained 4 percent, yet U.S. imports from China had already collapsed by 43 percent week-over-week in the preceding turmoil.¹³ Traditional risk-monitoring, derived from quarterly earnings releases, analyst reports, and end-of-day volatility indices, lags by days or weeks and misses the critical 1–3-day spikes in downside risk that policy reversals trigger. A real-time NLP approach on event descriptions closes this monitoring gap by delivering near-immediate signals of heightened tail-risk.

Hypothesis

With the observation that Trump's erratic behavior and rapid fire tariffs have been impacting the economy, my hypothesis is that negative, tech-focused trade policy in the U.S.- China trade war

⁵ Sterling, T. (2024, September 6). Dutch government retakes export control over two ASML tools from US. Reuters.
<https://www.reuters.com/technology/dutch-government-retakes-export-control-over-two-asml-tools-us-2024-09-06/>

⁶ Ibid.

⁷ US-China relations in the trump 2.0 ERA: A timeline - china briefing. China Briefing News. (2025, April 7).
<https://www.china-briefing.com/news/us-china-relations-in-the-trump-2-0-implications/>

⁸ Ibid.

⁹ Ibid.

¹⁰ Ibid.

¹¹ US and China agree to drastically roll back tariffs in major trade breakthrough. CNN. (2025, May 12).
<https://www.cnn.com/2025/05/12/business/us-china-trade-deal-announcement-intl-hnk>

¹² Wisnefski, S. (2025, April 3). Markets News, April 3, 2025: Trump tariffs spark worst day for major stock indexes since 2020; Dow loses almost 1,700 points, S&P drops nearly 5%. Investopedia.
<https://www.investopedia.com/dow-jones-today-04032025-11708250>

¹³ Carew, S., & Chauhan, S. (2025, May 12). Wall street equity indexes close higher after US-China Tariff Truce | Reuters. Reuters.
<https://www.reuters.com/world/china/us-stock-futures-jump-us-china-trade-stand-off-thaws-2025-05-12/>

drives higher downside risk for investors and firms in the technology sector. Specifically, trade policies that impact the supply chain or market for technology firms, as well as the frequency of these policies will drive up the value at risk for investors and firms.

To explore this hypothesis, I will be using a timeline dataset from the Peterson Institute for International Economics, which includes the date, category, and a text summary of the relevant trade event/policy.¹⁴ I will also be utilizing Yahoo finance data and Global Firepower Index data to add quantitative variables and targets to the dataset.

Exploratory Data Analysis

Before fully conducting exploratory data analysis (EDA), I had to apply natural language processing (NLP) techniques to my dataset. The original dataset had a column for the date, the country that the U.S. trade-oriented action was taken against, the category (trade deficits, trucks, autos, etc.), the description of the action/event, and several columns for links to information about the event. I subsetting the dataset to just the date, country, category, and description.

I continued to prepare my dataset by using VADER Sentiment analyzer to analyze the sentiment in the short descriptions of the trade events. It tends to do better with short pieces of text.¹⁵ Though this wasn't data parsed from social media, I still felt it was more robust than other sentiment analysis tools because it has a compound score for the entirety of the text that is being analyzed. The score is between -1 and 1.¹⁶ A score greater than 0.05 indicates positive sentiment.¹⁷ A negative score greater than -0.05 indicates negative sentiment, and a score between -0.05 and 0.05 indicates neutral sentiment.¹⁸ Using this, I assigned sentiment to each row (event) in the dataset and added it to the sentiment column.

Then, I used a keyword weighted TfidfVectorizer to assign a score to the event if it was related to the technology industry. The terms I elected to use were "semiconductor", "chip", "rare earth", "critical mineral", "electronics", "battery", "aerospace", "manufacturing", and "china". I preferred this method rather than manually cleaning the dataset to subset to just China or just tech industry events, because the dataset is very small (60 rows) and the presence of other trade-related events could still contribute to increased downside risk in the technology industry, if there are overall negative sentiments in the market. This way, the events related to technology trade are prioritized more heavily by the model.

¹⁴ Brown, C. P. (2025, April 29). Trump's Trade War Timeline 2.0: An up-to-date guide | piie. PIIE.
<https://www.piie.com/blogs/realtime-economics/2025/trumps-trade-war-timeline-20-date-guide>

¹⁵ GeeksforGeeks. (2025, May 10). Sentiment analysis using Vader - using Python.
<https://www.geeksforgeeks.org/python-sentiment-analysis-using-vader/>

¹⁶ Ibid.

¹⁷ Ibid.

¹⁸ Ibid.

Once the NLP process was done, I shifted to the financial data. In my original plan for this project, I broadly hypothesized that the desire to be ahead militarily is driving the trade war between the U.S. and China, especially with respect to semiconductors and technology that are used by the aerospace and defense industries. The best way to measure this was to measure impact on technology and highly innovative companies, which could be tracked in an aggregate way by using the NDX 100 index. The companies in the NDX index include Apple, Intel, Applied Materials, Microsoft, Adobe, Gilead, Microchip, Synopsys, Cisco, Qualcomm, ASML, Amazon, NVIDIA, AMD, Texas Instruments, Google, Tesla, Meta, and Palantir, among others (a mix of semiconductor, software, and defense contracting companies). I used the yfinance package to get price data for the time period from January 17, 2025 to May 15, 2025 (three days prior to the first day in the trade war dataset and three days after the last day in the trade war dataset).¹⁹ I then pulled the closing prices for each day, and computed the daily returns on the index.

After this, I computed the rolling value at risk (VAR). I chose value at risk because it quantifies the level of risk exposure that investors and firms care about when there are event that will impact market value.²⁰ By focusing on the 5% quantile of value at risk, I am only assessing the tail risk, or losses exceeded only 5% of the time, rather than average volatility.²¹ If I looked at average volatility, I don't think the model or data would be able to capture sharp downside spikes that are triggered in a contentious trade war such as this one between the U.S. and China. It also provides a clear, interpretable link between the political risk events I am analyzing through NLP and the magnitude of impact on financial loss in the market. Using a rolling-window construction makes it easier to align the model with event dates and assess how rapidly the market reprices risk in response to escalating tariffs and volatility.²² ²³ I used a 21 day window for the VAR since that represents about one business month or one month of trading days and appears to be standard in some VAR calculations in the risk industry.²⁴ I then converted the VAR I computed to positive to capture the magnitude of loss.

Next, I aggregated the events in the trade war data table by date, and got the average scores for the sentiment, events, and technology TF-IDF scores. I then added a 1-3 day rolling window for the text features to capture immediate market reactions and any lagged effects. I tested this later on by checking when the absolute correlation between VAR and sentiment peaks, which was at two days remaining consistent to the third day. I then merged this table with the VAR table and

¹⁹ Yfinance. PyPI. (n.d.). <https://pypi.org/project/yfinance/>

²⁰ Kenton, W. (2024, June 26). Understanding value at risk (VAR) and how it's computed. Investopedia. <https://www.investopedia.com/terms/v/var.asp>

²¹ Harper, D. R. (2024, June 4). What is value at risk (VAR) and how to calculate it?. Investopedia. <https://www.investopedia.com/articles/04/092904.asp>

²² Polyakova, A. (2024, April 15). Market risk analytics in python: Interactive rolling var. Atoti Community. <https://www.atoti.io/articles/market-risk-analytics-in-python-interactive-rolling-var/>

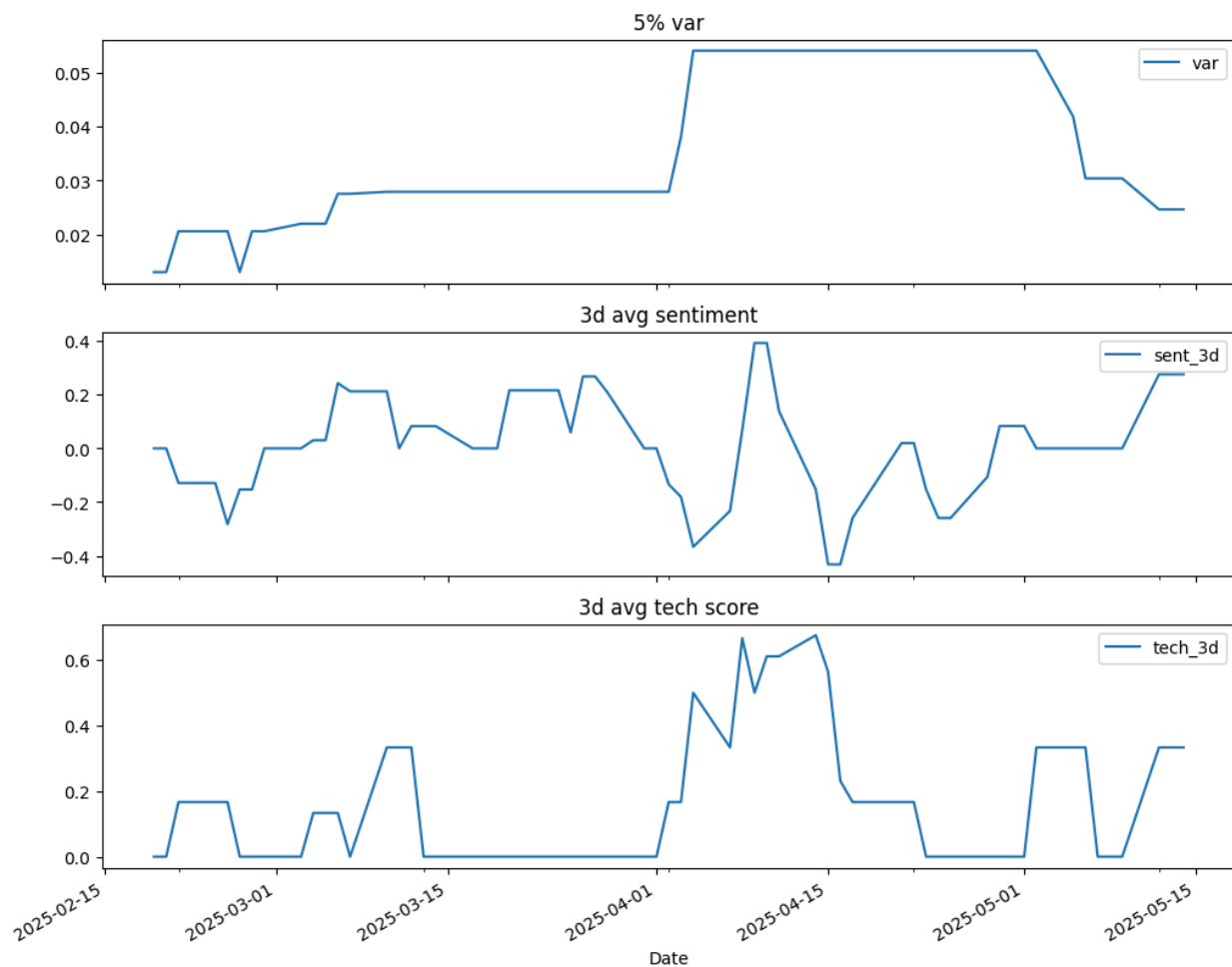
²³ Rolling Window Analysis. QuestDB. (n.d.). <https://questdb.com/glossary/rolling-window-analysis/>

²⁴ RiskSpan. (2020, January 28). Calculating var: A review of methods. <https://riskspan.com/calculating-var-a-review-of-methods/>

added a static ratio of China and the US 2025 Global Firepower Index. I used this because technology is highly leveraged by the U.S. and China for military purposes. This was an annually reported statistic, so I did not expect it to have any impact on the model's performance.²⁵

My final data table to be used for my models and EDA included the date, the calculated return, the VAR, the average sentiment for the day, the average technology score for the day, the average event count, and the GFI ratio.

I then checked the data for outliers and patterns in the data. The count_3d, which shows the count of events in a three day window, has a mean of 2.25 and a maximum value of 11, which shows 11 events in a three day period. I also checked the distribution of sent_3d and tech_3d. Those can take values between -1 and 1, so I was not as concerned about log-transforming those. Seeing that the count_3d had a heavy right skew, I log-transformed the data.



²⁵ 2025 military strength ranking. Global Firepower - World Military Strength. (n.d.).
<https://www.globalfirepower.com/countries-listing.php>

Figure 1: Time-Series Plot of VAR Risk Metric and NLP Features (3-Day Average Sentiment, 3-Day Average Tech Score)

I also plotted a time-series plot of the VAR risk metric and the sentiment feature and tech industry score. The top plot is for the 5% value at risk, which starts low through February and most of March (~1-2%). Around late March 2025, the VAR goes up to a little less than 3%, and jumps sharply around April 1 to about 5.5%, where it stays through mid-April before falling back towards ~2.5% by mid-May. This lines up with the market uncertainty and event timeline thus far, where China and the US escalated tariffs during the majority of the month of April, and reached a tentative 90 day agreement on May 12.²⁶

The three-day average sentiment plot shows that sentiment is slightly above and below zero in March, with some negative dips. Before the big jump in VAR, there was a spike to strongly positive sentiment followed by a dip to about -0.4 in mid-April. The volatility in sentiment derived by the VADER Sentiment model lines up with the peak in downside risk in the VAR plot.

The three day average tech-relevance event score from TF-IDF shows that tech-related events in the trade war were around zero in March, but the score shoots up around late March/early April, to around 0.7. This likely reflects the cluster of semiconductor and rare-earth export reports in the timeline. They taper off after mid-April, and the tech score declines back toward zero, just as the VAR is also declining.

The event-driven tech tariffs and trade restrictions are captured in a big spike in the tech_3d feature. The sentiment around those days was weirdly positive, to strongly negative (likely because of retaliation), and the combined effect seems to have pushed the market's 5% VAR higher. The downside risk receded when the tech-specific and negative sentiment events subsided.

This visual lends confidence to the idea that the combination of sentiment and tech relevance tracks movement in downside risk, which we can be quantified in an ML model.

²⁶ US and China agree to drastically roll back tariffs in major trade breakthrough. CNN. (2025, May 12). <https://www.cnn.com/2025/05/12/business/us-china-trade-deal-announcement-intl-hnk>

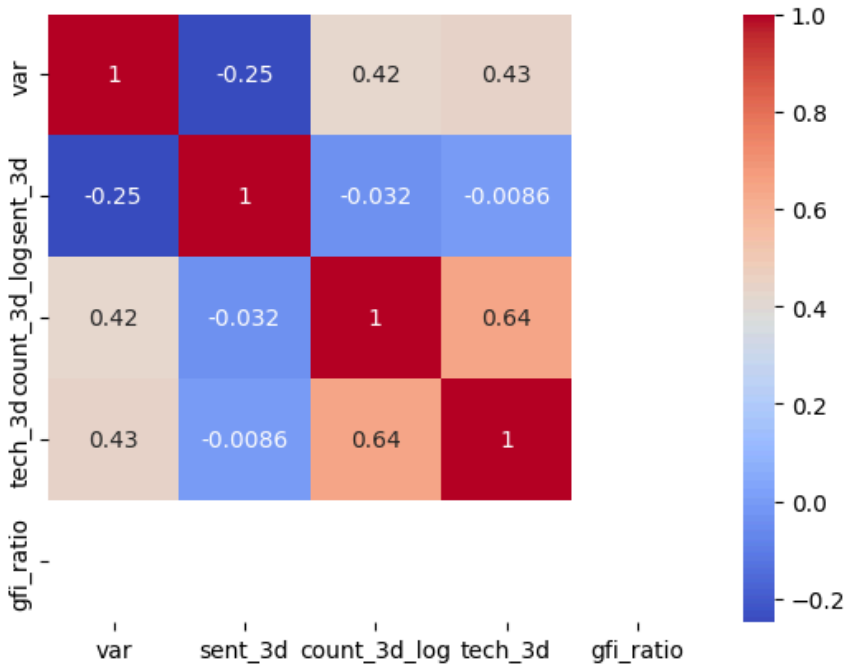


Figure 2: Correlation Matrix of Features in Dataset

The correlation matrix shows the following:

- var and sent_3d: This has a Pearson score of -0.25, showing that more positive sentiment leads to slightly lower downside risk. This intuitively makes sense.
- var and count_3d_log: This has a score of 0.42, which is moderately positive. This implies that days with more events tend to have higher var.
- var and tech_3d: This is 0.43, also moderately positive. This shows that days with more tech-centered trade policies line up with higher value at risk in the NDX 100. It makes sense that there would be greater potential for financial losses if there are more tech centered trade policies in the trade war.
- sent_3d and count_3d_log: This is close to 0 (0.03) which means that sentiment strength is pretty much independent of how many events occur. This makes sense, given that it is the nature of those events that determine sentiment.
- sent_3d and tech_3d: This is -0.01, which is also close to 0, showing that the sentiment isn't simply driven by the relationship of an event to the technology sector.
- count_3d_log and tech_3d: This has a score of 0.64, which implies that where there are clusters of more events, they tend to have an impact on the tech industry. This also shows mild collinearity, which could interfere with a parametric model but would probably be fine for a Random Forest model or a gradient boosted model.

Prototype Model and Resulting Time-Series Plot

For this task, which is predicting a 5% rolling value at risk on the Nasdaq-100 from text-derived signals, I considered a Random Forest Model (RF) and Gradient Boosting Regressor Model (GB). The Random Forest relies on bootstrap sampling and averages the predictions of the samples, which could be good for the spikes in event counts and tech scores and the target variable 'var', which has discrete jumps. Gradient Boosting builds the trees sequentially and corrects previous iterations of the ensemble, and is better for small datasets (I only have 60 rows/events).

Recall the hypothesis: does negative, tech-focused trade policy in the U.S.- China trade war drive higher downside risk? The hypothesis is looking for a causal link between sentiment/tech focus and downside risk. This requires supervised learning as an approach, because VAR is modeled as the outcome, and we can quantify feature importances and get interpretable effect sizes such as how a one unit decrease in sentiment drives up value at risk. An unsupervised method could produce a risk index by using the four features of tech_3d, sent_3d, count_3d_log, and gfi_ratio and aggregating them into a single score, but that misses the bulk of why I did the NLP, which is to see how much each NLP feature predicts the market's downside risk. I utilized both models and assessed their performance.

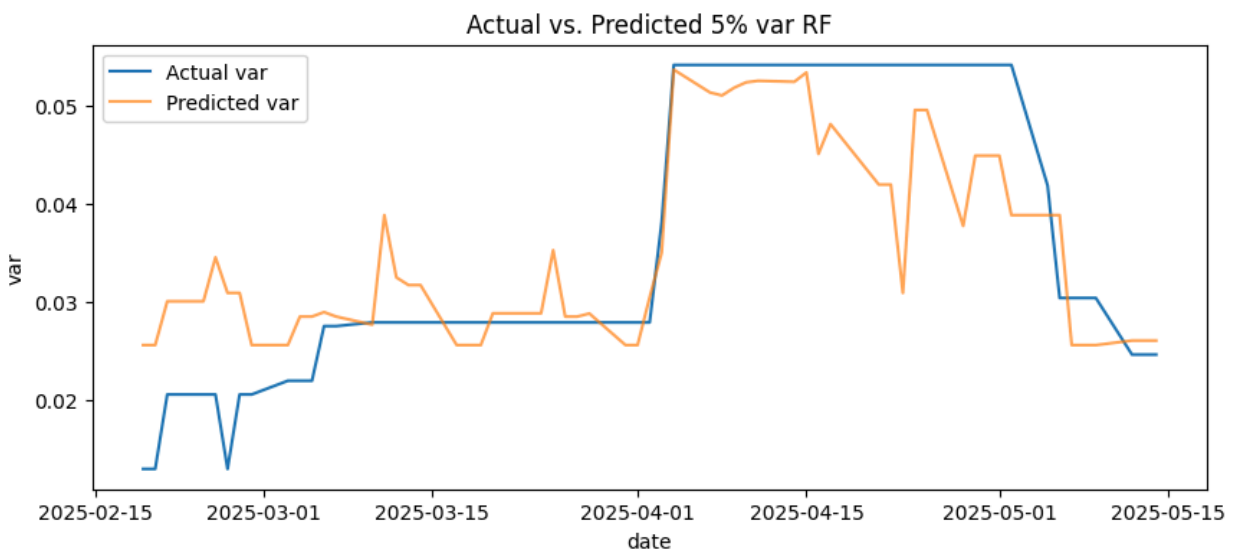


Figure 3.1: Actual versus Predicted 5% VAR (Random Forest Model)

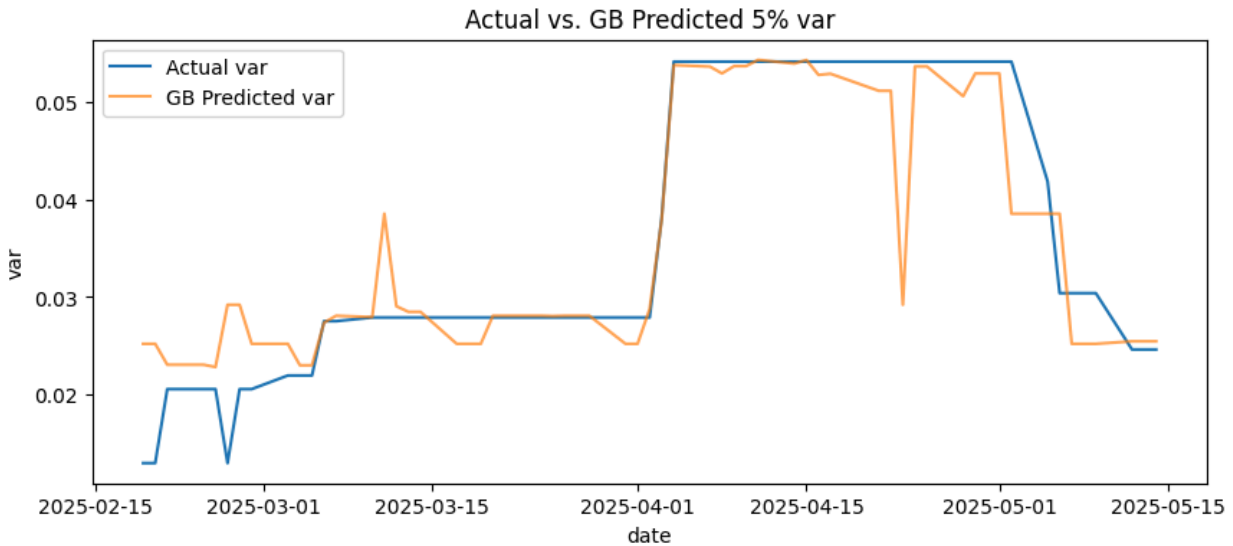


Figure 3.2: Actual versus Predicted 5% VAR (Gradient Boosted Model)

I used time-series cross-validation to assess how well my models generalize to unseen market conditions without look-ahead bias.^{27 28} Look-ahead bias in this context refers to relying on data that wasn't available during the period being studied. By splitting the data into sequential train and test folds, I made sure that each test set only contained dates that occurred after its corresponding training window. This approach also allowed me to compute out-of-sample metrics (MSE and R2) across multiple temporal folds, giving a more reliable picture of model stability and helping guard against overfitting on any single time period to the best of my ability.

The GB R2= -5.96 and the RF R2= -5.62. These are really high values for the R2 score. On average, the Gradient Boosting model's squared errors are 5.96 times larger than the total variance of the 'var' target variable, and the RF model's squared errors are 5.62 times larger than the total variance of the 'var' target variable. This means that the model is doing much worse than just predicting the mean value at risk every day. The Random Forest has a smaller MSE and a smaller R2, but the GB model explains more of the day-to-day variance of the var than the RF model.

The large R2 values could indicate a few things stemming from the data that are causing this outcome. The target variable of 'var' is very noisy. I rolled the 5% value at risk on a 21 day window initially, and this means that it can jump around a lot and make it hard to predict just

²⁷ Look-ahead bias. Corporate Finance Institute. (2024, February 15).

<https://corporatefinanceinstitute.com/resources/career-map/sell-side/capital-markets/look-ahead-bias/#:~:text=Time%20%20minutes-,What%20is%20Look%20Dahead%20Bias%3F,from%20a%20study%20or%20simulation.>

²⁸ GeeksforGeeks. (2025a, April 28). Time series cross-validation.

<https://www.geeksforgeeks.org/time-series-cross-validation/>

from three-day sentiment and event features we derived from the NLP. I also might have insufficient features. There are four predictors, one of which is constant for China and for the US (the global firepower index value) and three NLP derived features from a relatively small dataset (only 61 rows). These aren't sufficient predictors for a target variable that is probably driven by lots of other market forces (employment, earnings, trade flows, and Fed policy, among other things). The small dataset also means the model could be overfitting to patterns and isn't generalizing properly.

The plots for the GB and RF models basically capture the same broad trends, such as low VAR in February through March, a sharp jump in early April, and a decline in early May. Notably, both models predict more variation between early April and early May, while the actual VAR plateaus during this time.

The GB predictions are smoother than the RF. The RF model overestimates downside risk in early March and underestimates it during the April-May plateau. The GB tracks the actual var predictions more tightly, especially around the high risk plateau, with some under and over shoots around mid-April. The closer fit is reflected in the higher cross validated R2 score for the GB model. Both the RF and GB models perform worse early on in the data, but this is likely due to the sparsity in the original NLP dataset of events that took place prior to April, when the trade war really picked up speed. The large dip both models have during the plateau suggest overfitting to training data. The gradient-boosted model seems to align better with realized risk and appears more precise.

The feature importances are very similar for both models, with `sent_3d` being most important in both models (higher score in the RF model) and `tech_3d` being second most important (GB has a higher score than the RF model for this). GFI ratio has an importance of zero. The `count_3d_log` is the third most important feature, with the RF model having a higher feature importance score than the GB model. This makes sense, given that the sentiment is driving the market, as well as the importance of technology-related trade driving the downside risk in the NDX index specifically.

To further assess the performance of the models, I compared the residuals and SHAP values of the models, which also showed that the Gradient Boosted model had a much tighter fit, and that sentiment, tech focus, and event count were driving the model's predictions. Both models underpredicted around mid-April, when the trade war escalated. The RF model underestimates on actual high-risk days.

My final result was the following time-series plot from the GB model, which has technology industry related trade events annotated with dashed lines. Where there are higher concentrations

of technology related events, there are more peaks and declines predicted by the GB model, influenced by the sentiment score and technology impact of the event.

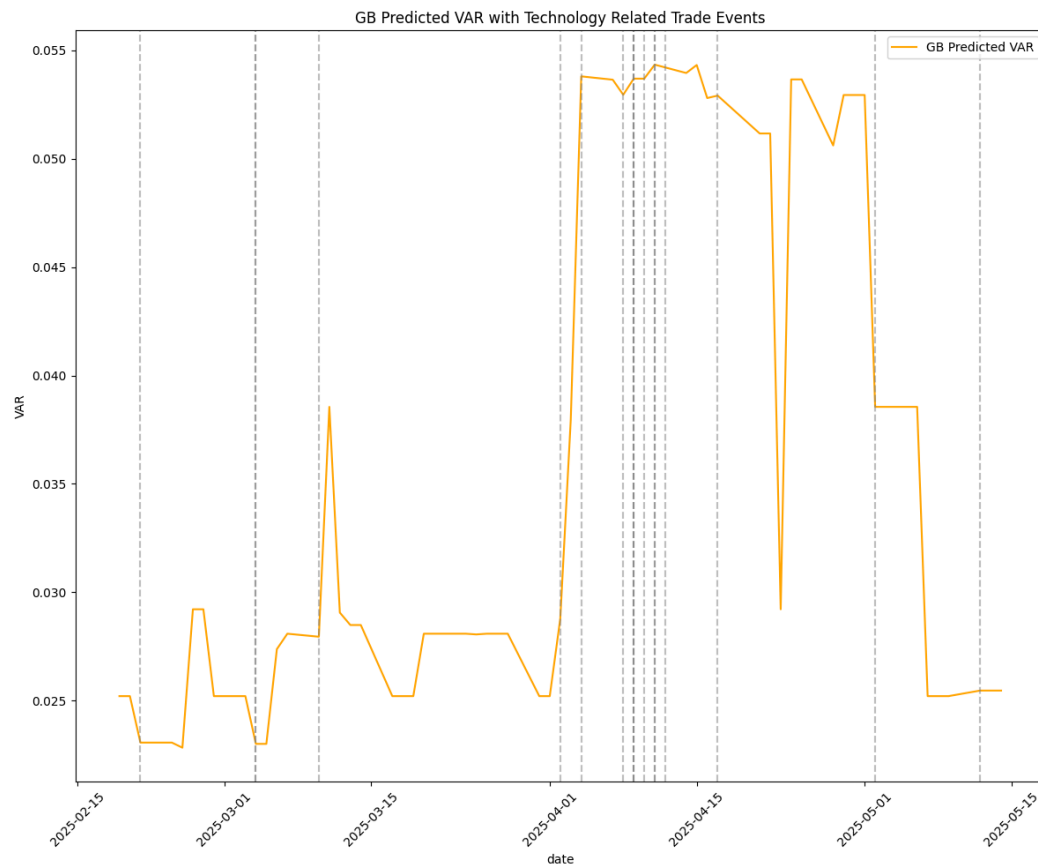


Figure 4: Gradient Boosted Regression Model Predicted VAR with Annotated Tech Trade Events

Risk Analysis

Both models flagged the early-April spike in the U.S. China tensions, when export restrictions on rare-earth metals and retaliatory U.S. tariffs hit the market, as a key moment of high predicted downside risk. The 5% VAR jumped again when China announced rare-earth export controls and again when reciprocal tariffs were raised to 125%.^{29 30} The GB model did over and under predict around the policy inflection points, but the GB model did track the sustained high-risk plateau

²⁹Baskaran, G., & Schwartz, M. (2025, April 14). The consequences of China's New Rare Earths Export Restrictions. CSIS. <https://www.csis.org/analysis/consequences-chinas-new-rare-earths-export-restrictions>

³⁰ The United States Government. (2025, April 10). Modifying reciprocal tariff rates to reflect trading partner retaliation and alignment. The White House. <https://www.whitehouse.gov/presidential-actions/2025/04/modifying-reciprocal-tariff-rates-to-reflect-trading-partner-retaliation-and-alignment/>

through mid-April more tightly. Its performance when analyzing residuals was also an improvement over the RF model.

The pre-registered hypothesis, that negative, tech-focused trade policy drives higher downside risk for technology firms, was qualitatively supported. Both `tech_3d` and `count_3d_log` had strong positive SHAP and feature importance signals, showing that clusters of technology related trade events drive up the VAR. '`sent_3d`' provided directional counterbalance, in that negative sentiment increased risk, and positive sentiment reduced risk. The negative R2 however indicates that they only explain part of the total VAR variance, even though they are important.

Some social science assumptions we had to make for this were that the compound sentiment analysis score provided by VADER accurately captures market-relevant sentiment in short policy descriptions. This might miss nuance in formal event text since it's primarily designed for social media text analysis. I am also relying on the TF-IDF lexicon I manually created for keyword weighting as sufficiently isolating the technology industry component of the trade war. Other terms that might be tangentially related to this component, such as shipping or ports may have been missed. This might not capture the effect on the supply chain that impacts technology, especially if other policies mentioned were so broad they were not captured by the lexicon. I also had to assume that I was using enough variables for market prediction (NLP to capture fear, which can drive markets) and using a good stock index to create my target variable.

The market and commercial implications here are that a risk manager for a firm or investment bank with business interests in technology and defense firms could track the three-day event index and sentiment score to anticipate VAR jumps, rather than waiting for realized volatility to spike. Firms with large exposure to these policies could also change their hedging strategies if the model flags a cluster of tech-oriented trade movements and negative-sentiment events. However, rather than directing what these firms in the NDX index should do, this gives more utility to people investing in those firms, which means that this model isn't really helpful to the client base I envisioned initially.

Clients (investors, firm stakeholders) can use the prototype model to monitor news feeds or headlines, so that alerts are triggered when `tech_3d` or `sent_3d` pass certain thresholds that would indicate a next-day VAR spike. Investors can also adjust their exposure to technology and defense-tech companies when the VAR exceeds some threshold instead of reacting after the losses occur.

Product Development Steps

The largest constraint for the supervised models were a small sample size, a noisy target variable, and only four predictors, one of which had no impact on the model's performance. To

improve the Gradient Boosted model's predictive power, I would first change my approach to sentiment analysis and natural language processing. I would consider replacing VADER with another type of transformer model, and also using LDA to extract relevant topics that would give the model more context beyond the self-constructed keywords for TF-IDF. I might also consider adding in trading volume, trade flows of goods, USITC data, and industry data such as employment data to improve the model's explanatory abilities and account for other factors that could be driving the results we see in the actual VAR. I would also hyperparameter tune my Gradient Boosting model using GridSearchCV to ensure that the model is using the best possible parameters for prediction. I would also increase the sample size. This could be done by collecting more data from places like NewsAPI, or also conducting the same analysis with other geopolitical pairs (such as the U.S. and Canada, or some other pair) to reduce overfitting and give the model more example of places where there is tail-risk. I could also choose more granular firm-level target variables and pick more firm-level data to conduct NLP on.